

CS201 Midterm 3

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Abstract—Write only ONE function per question. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures including arrays, (iv) you can use any function given in the data structure, (v) you can not modify the data structure contents.

I. QUESTION (HEAP)

Write the method in **MaxHeap** class

```
int shortestDistanceBetweenNodes(
    int index1,
    int index2)
```

that returns the shortest distance between two nodes in the heap with indexes index1 and index2. Generate the ascendant lists of the nodes with index1 and index2 (You can assume the tree depth is smaller than 100). Compare those lists to solve the problem.

II. QUESTION (HEAP)

Write the method in **MinDHeap** class

```
int sumOfMaxChange(int min, int max)
```

that returns the sum of the lengths of the change intervals of each node. A change interval of a node N is defined with the minimum and maximum values that can be attained by N without changing any other node. For root node, take the minimum as min . For leaf nodes, take the maximum as max .

III. QUESTION (DISJOINT SET)

Given the index of a set, write a function that returns the ancestors (itself, parent, grandparent, etc.).

```
int* ascendants(int index)
```

The size of the returning array should be as much as needed.

IV. QUESTION (DISJOINT SET)

Write a method

```
bool isValid()
```

that returns true when the given disjoint set is valid, that is from every node n , when the ascendants are traversed, no circularity is observed (that is you do not encounter the node n again).

V. QUESTION (GRAPH)

For a directed weighted graph, write the method in linked list implementation

```
int shortestIn2Hops(int index1, int index2)
```

which returns the shortest distance between the nodes index1 and index2 by two hops, that is, it will return of the shortest of all paths, where one goes from index1 node to node i , then from node i to node $index2$.

VI. QUESTION (GRAPH)

For a directed weighted graph, write the method in array implementation

```
int lengthOfCircle()
```

which returns length of the circle assuming that the graph is a circular graph. A graph is circular if all the nodes create a circular path. Each node is connected to two others, like points on a circle.